

Information Gathering (Reconnaissance)

Objective:

The objective of the reconnaissance phase is to collect basic information about the target system without exploiting any vulnerabilities. This phase helps understand what services exist, where they are exposed, and how the target system is reachable.

Environment Setup:

- Attacker Machine: Kali Linux
- Target Machine: Ubuntu Linux running DVWA
- Network Type: Internal network

Types of Reconnaissance:

- Passive Reconnaissance
 - Collects information without modifying or attacking the target
 - Minimal interaction
 - No exploitation
- Active Reconnaissance
 - Direct interaction with the target
 - Still non-exploitative
 - Used to confirm availability and behaviour

PASSIVE RECONNAISSANCE:

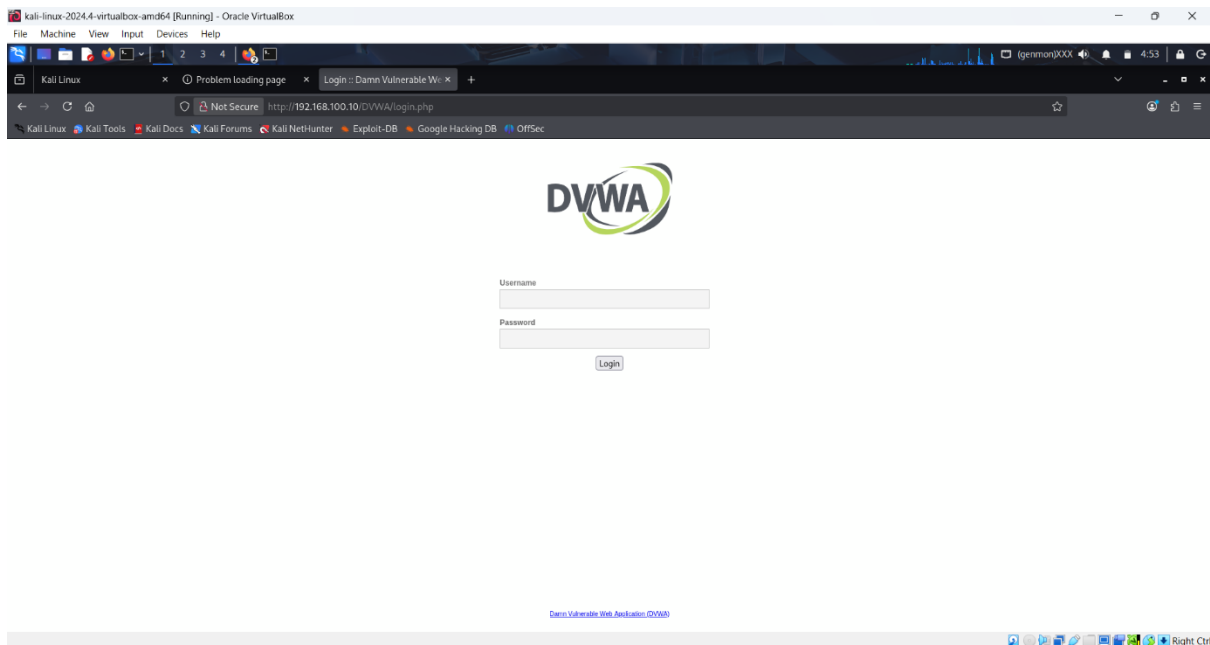
(1) Manual Browser Reconnaissance:

- Tool Used:
Web Browser (Firefox in Kali Linux)
- This Tool Is Used To:
 - Confirm the target system is alive
 - Identify the running web application (DVWA)
- It Reveals:
 - URL structure
 - Input forms

- Cookies
- Basic HTTP behavior

- Action Performed:

`http://192.168.100.10/DVWA/`



DVWA login page loads successfully and URL bar showing Ubuntu IP address

ACTIVE RECONNAISSANCE:

(2) Ping:

- Tool Used

ping

- Command

`ping 192.168.100.10`

- This command Is Used To:

- Confirm that the target system is reachable
- Verify network connectivity
- Measure basic response time

```
kali@kali: ~  
$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel  
    state UP group default qlen 1000  
    link/ether 08:00:27:6e:13:6e brd ff:ff:ff:ff:ff:ff  
    inet 192.168.100.20/24 brd 192.168.100.255 scope global nopref  
ixroute eth0  
    valid_lft forever preferred_lft forever  
    inet6 fe80::d660:bc7a:374d:2cd2/64 scope link noprefixroute  
    valid_lft forever preferred_lft forever  
  
(kali@kali)~  
$ ping 192.168.100.10  
PING 192.168.100.10 (192.168.100.10) 56(84) bytes of data:  
64 bytes from 192.168.100.10: icmp_seq=1 ttl=64 time=1.45 ms  
64 bytes from 192.168.100.10: icmp_seq=2 ttl=64 time=1.82 ms  
64 bytes from 192.168.100.10: icmp_seq=3 ttl=64 time=2.18 ms  
64 bytes from 192.168.100.10: icmp_seq=4 ttl=64 time=8.46 ms  
^Z  
zsh: suspended ping 192.168.100.10  
  
(kali@kali)~  
$
```

Successful ICMP ping responses

(3) CURL:

curl is a command-line tool used to:

- Send HTTP requests
- Retrieve server responses
- Analyze headers and content

It is commonly used in both passive and active reconnaissance.

curl helps to identify web server type, HTTP status codes, cookies, application behaviour.

It can reveal information leakage, Misconfigurations, Server details.

A. Fetch Web Page Content

Command:

```
curl http://192.168.100.10/dvwa/
```

```

zsh: suspended ping 192.168.100.10

(kali@kali)-[~]
$ curl http://192.168.100.10/DVWA/login.php
<!DOCTYPE html>

<html lang="en-GB">

  <head>

    <meta http-equiv="Content-Type" content="text/html
; charset=UTF-8" />

    <title>Login :: Damn Vulnerable Web Application (D
VWA)</title>

    <link rel="stylesheet" type="text/css" href="dvwa/
css/login.css" />

  </head>

  <body>

    <div id="wrapper">

      <div id="header">

        <br />

```

Raw HTML source code of the DVWA login page

B. Fetch HTTP Headers Only

Command

```
curl -I http://192.168.100.10/dvwa/
```

```

    <p><a href="https://github.com/digininja/DVWA/" target="_b
lank">Damn Vulnerable Web Application (DVWA)</a></p>

    </div> <!--<div id="footer"> -->

    </div> <!--<div id="wrapper"> -->

    </body>

</html>

(kali@kali)-[~]
$ curl -I http://192.168.100.10/DVWA/
HTTP/1.1 302 Found
Date: Sun, 04 Jan 2026 12:41:08 GMT
Server: Apache/2.4.58 (Ubuntu)
Set-Cookie: security=low; path=/
Set-Cookie: PHPSESSID=qshkmsae9e11eijkt19hav7ehd; expires=Mon, 05
Jan 2026 12:41:08 GMT; Max-Age=86400; path=/
Expires: Thu, 19 Nov 1981 08:52:00 GMT
Cache-Control: no-store, no-cache, must-revalidate
Pragma: no-cache
Location: login.php
Content-Type: text/html; charset=UTF-8

```

Terminal showing HTTP response headers

Importance Of Reconnaissance:

- Identifies live systems and exposed services
- Helps understand the attack surface
- Reduces blind or random attacks
- Forms the foundation for scanning and exploitation
- Allows attackers to plan efficient attack strategies

OWASP TOP 10 MAPPING:

A02:2025 – Security Misconfiguration

- DVWA application is directly accessible via IP address
- Server type and OS information disclosed through HTTP headers
- No access restrictions at reconnaissance stage

A09:2025 – Security Logging & Alerting Failures

- Reconnaissance activities such as browser access, ping, curl requests were not detected or logged
- No alerts or warnings triggered during information gathering

Conclusion:

The reconnaissance phase successfully identified the target system, confirmed network connectivity, and revealed the presence of the DVWA web application. Information such as server type, operating system, session cookies, and application structure was collected without exploiting any vulnerabilities.